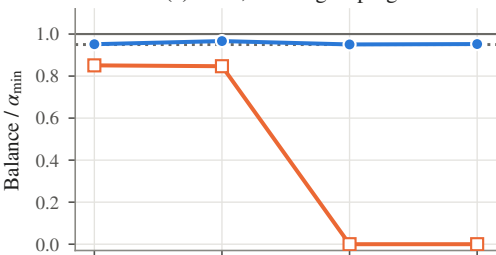


—●— Fair ParPIC (ours)
 —□— ParPIC + k -means
 Floor $(1 - \delta)\alpha_{\min}$, $\delta = 0.05$
 — Ceiling $\alpha_{\min}$

(a) Race, nested groupings



(b) Different sensitive attributes



—●— Fair ParPIC (ours)

- - - □ - - - ParPIC + fair assignment

